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18.02SC

Multivariable Calculus 18.02SC — Multivariable Calculus May 2026 **EXTREMELY****HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping with contraction factor $k < 1$. Prove the Banach fixed-point theorem: T has a unique fixed point. Additionally, derive the error estimate $\|x_n - x^*\| \leq \frac{k^n}{1-k} \|x_1 - x_0\|$ where $x_{n+1} = T(x_n)$.
2. (3 points) Analyze the stability of the Runge-Kutta method of order 4 when applied to the stiff ODE $y' = \lambda y$ with $\lambda \in \mathbb{C}$, $\operatorname{Re}(\lambda) < 0$. Determine the region of absolute stability and discuss how the step size h must be chosen for stiff problems.
3. (5 points) Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of functions of bounded variation with $\operatorname{Var}(f_n) \leq C$ uniformly and $f_n \rightarrow f$ pointwise. Prove that f is of bounded variation and $\operatorname{Var}(f) \leq C$. Give a counterexample showing that uniform bounded variation is necessary.
4. (4 points) State and prove the Implicit Function Theorem. Apply it to show that the equation $F(x, y) = y^5 + y + x = 0$ can be solved for $y = g(x)$ near $(0, 0)$, and compute $g'(0)$.
5. (7 points) Classify the second-order linear PDE $u_{xx} + 2u_{xy} + u_{yy} + u_x + u_y = 0$. Determine its type (elliptic, parabolic, hyperbolic) and find its canonical form. Solve the resulting canonical equation.
6. (4 points) Prove the Rellich-Kondrachov compactness theorem: $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ compactly for $1 \leq q < p^*$ when Ω is bounded and $\partial\Omega$ is C^1 .
7. (5 points) Determine all values of $p \in \mathbb{R}$ for which the series $\sum_{n=2}^{\infty} \frac{1}{n^p \ln n}$ converges. Prove your answer using the integral test and comparison tests.
8. (3 points) Prove that $\mathbb{R}P^3$ is diffeomorphic to $SO(3)$ by constructing explicit maps and verifying they are smooth inverses. Use this to compute $\pi_1(SO(3))$.
9. (3 points) Prove using the epsilon-delta definition that the function $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$ is differentiable at $x = 0$ but $f'(x)$ is not continuous at $x = 0$. Carefully justify all steps in your epsilon-delta arguments.

10. (2 points) Prove the Schauder fixed-point theorem using Schauder estimates and the Arzela-Ascoli theorem: a compact operator $T : X \rightarrow X$ on a Banach space has a fixed point if it maps a closed convex set into itself.
11. (3 points) Prove the Hopf-Rinow theorem: for a Riemannian manifold (M, g) , metric completeness is equivalent to geodesic completeness (every geodesic extends for all time).
12. (2 points) Prove the Atiyah-Singer index theorem for a simple case: compute the index of the Dirac operator on S^2 and verify it equals the integral of the $\hat{A}$ -genus.
13. (5 points) Prove the Nash embedding theorem for compact Riemannian manifolds: any compact C^∞ Riemannian manifold can be isometrically embedded into $\mathbb{R}^N$ for sufficiently large N .
14. (4 points) Let M be a compact oriented n -manifold with boundary ∂M . State and prove Stokes' theorem for differential forms. Then apply it to prove the divergence theorem: $\int_M \operatorname{div} F \, dV = \int_{\partial M} F \cdot n \, dS$.
15. (6 points) Prove the Gagliardo-Nirenberg-Sobolev inequality: for $1 \leq p < n$, there exists $C = C(n, p) > 0$ such that $\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}$ for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = np/(n - p)$.
16. (5 points) Prove the Sobolev embedding theorem: for $p > n$, $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\alpha}(\mathbb{R}^n)$ with $\alpha = 1 - n/p$. In particular, show that if $u \in W^{1,p}(\mathbb{R}^n)$ with $p > n$, then u is (almost everywhere equal to) a Hölder continuous function.
17. (2 points) Prove the maximum principle for elliptic PDEs: if $Lu = -\Delta u + c(x)u \geq 0$ in Ω with $c \geq 0$, then $\max_{\bar{\Omega}} u \leq \max_{\partial\Omega} u^+$. Show how this implies uniqueness for the Dirichlet problem.
18. (3 points) Prove Sard's theorem: if $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is C^k with $k > \max(n - m, 0)$, then the set of critical values of f has Lebesgue measure zero in $\mathbb{R}^m$.
19. (5 points) Prove the Brouwer fixed-point theorem using differential topology: any continuous map $f : \bar{B}^n \rightarrow \bar{B}^n$ has a fixed point. Use the fact that there is no smooth retraction of the ball onto its boundary.
20. (4 points) Prove the Lax-Milgram theorem and use it to show existence and uniqueness of weak solutions to the Dirichlet problem $-\Delta u = f$ in Ω , $u|_{\partial\Omega} = 0$ for $f \in L^2(\Omega)$.
21. (4 points) Prove that symplectic integrators preserve the symplectic structure $\omega = \sum dq^i \wedge dp_i$ of Hamiltonian systems. Show that the Verlet (Störmer) method is symplectic and has energy error bounded over exponentially long times for integrable systems.

22. (5 points) Prove the Hille-Yosida theorem: a linear operator A generates a strongly continuous contraction semigroup on a Banach space X iff A is closed, densely defined, and all $\lambda > 0$ are in the resolvent set with $\|R(\lambda, A)\| \leq 1/\lambda$.
23. (3 points) Prove that the Laplace operator Δ on a compact Riemannian manifold has discrete spectrum $\lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$ and each eigenvalue has finite multiplicity. Show $\lambda_1 = \inf \frac{\|\nabla u\|^2}{\|u\|^2}$ over nonzero $u \in H^1(M)$ with $\int u = 0$.
24. (6 points) Prove that there is no continuous linear functional $L : \ell^\infty \rightarrow \mathbb{R}$ that extends the limit functional on c (convergent sequences) and satisfies $\|L\| = 1$. (Hint: use the Hahn-Banach theorem and consider the Banach limit.)
25. (3 points) Prove the Divergence theorem for smooth vector fields on compact manifolds with boundary and derive the Green's identities.

Solutions

Solution 1:

Pick any $x_0 \in X$, define $x_{n+1} = T(x_n)$. Then $d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)$. By triangle inequality, $d(x_{n+m}, x_n) \leq \sum_{i=0}^{m-1} k^{n+i} d(x_1, x_0) \leq \frac{k^n}{1-k} d(x_1, x_0)$. So $\{x_n\}$ is Cauchy, converges to x^* . By continuity of T , $T(x^*) = x^*$. Uniqueness: if $Tx = x$, $Ty = y$, then $d(x, y) = d(Tx, Ty) \leq kd(x, y)$, so $d(x, y) = 0$. The error estimate follows from taking $m \rightarrow \infty$.

Solution 2:

The RK4 method applied to $y' = \lambda y$ gives $y_{n+1} = R(z)y_n$ where $z = h\lambda$ and $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$. The stability region is $\{z \in \mathbb{C} : |R(z)| \leq 1\}$. For stiff problems where $\text{Re}(\lambda)$ is large negative, $|R(z)|$ grows for large $|z|$ (RK4 is not A-stable), requiring h small enough that $h\lambda$ stays in the stability region.

Solution 3:

For any partition $0 = x_0 < \dots < x_k = 1$, $|f(x_i) - f(x_{i-1})| = \lim_n |f_n(x_i) - f_n(x_{i-1})| \leq \liminf_n \text{Var}(f_n) \leq C$. So $\text{Var}(f) \leq C$. Counterexample without UB: $f_n(x) = nx$ on $[0, 1/n]$, $f_n(x) = 1$ elsewhere. $\text{Var}(f_n) = 2n$, $f_n \rightarrow 1$ pointwise, but limit has variation 0, so inequality fails direction.

Solution 4:

IFT: Let $F : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$ be C^1 , $F(a, b) = 0$, and $\partial F / \partial y(a, b)$ invertible. Then there exists $g : U \rightarrow \mathbb{R}^n$ C^1 near a with $F(x, g(x)) = 0$ and $Dg(x) = -(\partial F / \partial y)^{-1} \partial F / \partial x$. For $F(x, y) = y^5 + y + x$, $\partial F / \partial y = 5y^4 + 1$, at $(0, 0)$ this is $1 \neq 0$, so IFT applies. $g'(0) = -(\partial F / \partial y)^{-1} \partial F / \partial x = -1/1 = -1$.

Solution 5:

The characteristic equation: $a_{11}dy^2 - 2a_{12}dxdy + a_{22}dx^2 = 0$ gives $1 \cdot dy^2 - 2 \cdot 1 \cdot dxdy + 1 \cdot dx^2 = 0$, so $(dy - dx)^2 = 0$. Double root: parabolic type. Characteristic curves: $y - x = c$. Canonical coordinates: $\xi = y - x$, $\eta = x$. Then $u_{\eta\eta} + u_\eta = 0$. Solve: $u(\xi, \eta) = A(\xi)e^{-\eta} + B(\xi)$, i.e. $u(x, y) = A(y - x)e^{-x} + B(y - x)$.

Solution 6:

Take a bounded sequence $\{u_n\}$ in $W^{1,p}$. By the Sobolev-Gagliardo-Nirenberg inequality, it's bounded in L^{p^*} . Use the Frechet-Kolmogorov criterion for compactness in L^q : show $\|u_n(\cdot + h) - u_n\|_{L^q} \rightarrow 0$ uniformly as $h \rightarrow 0$ using difference quotients and the $W^{1,p}$ bound. Extract a subsequence convergent in L^q by the Arzela-Ascoli type argument.

Solution 7:

By the integral test, $\int_2^\infty \frac{dx}{x^p \ln x}$ converges iff $p > 1$. For $p = 1$, $\int_2^\infty \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} \ln(\ln b) - \ln(\ln 2) = \infty$, so diverges. For $p < 1$, compare with $1/x^p$: $\frac{1}{n^p \ln n} > \frac{1}{n^p \cdot n^\epsilon}$ for large n , which diverges by p-test. For $p > 1$, $\frac{1}{n^p \ln n} < \frac{1}{n^{(p+1)/2}}$ for large n , which converges.

Solution 8:

Map $\mathbb{RP}^3 \rightarrow SO(3)$: represent a point in $\mathbb{RP}^3$ as a unit quaternion q modulo sign. The conjugation map $v \mapsto qvq^{-1}$ on pure quaternions gives a rotation in $SO(3)$. This is a smooth double cover $S^3 \rightarrow SO(3)$ with kernel $\{\pm 1\}$, inducing $\mathbb{RP}^3 \cong SO(3)$. Thus

$$\pi_1(SO(3)) \cong \mathbb{Z}/2\mathbb{Z}.$$

Solution 9:

We prove $f'(0) = 0$ by showing $|(f(h) - f(0))/h - 0| = |h \sin(1/h)| \leq |h|$. For any $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $|h| < \delta$ implies $|h \sin(1/h)| < \varepsilon$, so $f'(0) = 0$. For $x \neq 0$, $f'(x) = 2x \sin(1/x) - \cos(1/x)$. This oscillates wildly near 0 since $\cos(1/x)$ has no limit. Hence f' is not continuous at 0.

Solution 10:

Let $C \subset X$ be closed convex, $T(C) \subset C$, T compact. Approximate T by finite-rank operators T_n using Schauder basis. By Brouwer, each T_n has a fixed point x_n . By compactness of T , $\{T(x_n)\}$ has convergent subsequence $T(x_{n_k}) \rightarrow y$. Since $x_{n_k} = T_{n_k}(x_{n_k})$ and $T_{n_k} \rightarrow T$ uniformly, y is a fixed point of T .

Solution 11:

($\Rightarrow$): Assume metric completeness. For a geodesic γ with maximal interval (a, b) , if $b < \infty$, take $t_n \rightarrow b$. Then $\{\gamma(t_n)\}$ is Cauchy (distance $\leq C|t_n - t_m|$) so converges. Extend γ using the exponential map at the limit point. ($\Leftarrow$): Assume geodesic completeness. Take a Cauchy sequence p_n . Connect each p_n to p_0 by a minimizing geodesic. Bound on distances and completeness of initial data gives convergent subsequence.

Solution 12:

The Dirac operator D on S^2 acts on spinors. $\text{ind}(D) = \dim \ker D - \dim \ker D^*$. By Lichnerowicz formula, $D^2 = \nabla^* \nabla + \frac{1}{4}R$. On S^2 with $R = 2$, $\ker D$ consists of the constant spinor. So $\text{ind}(D) = 1$. The $\hat{A}$ -genus: $\int_{S^2} \hat{A}(TS^2) = \int_{S^2} 1 - \frac{p_1}{24} + \cdots = 1$ since $p_1(TS^2) = 2\chi = 4$ but the integral gives 0 at this order. Indeed $\int_{S^2} \frac{R}{4\pi} = \chi/2 = 1$.

Solution 13:

Sketch: Use the implicit function theorem to construct a short immersion $f_0 : M \rightarrow \mathbb{R}^N$. The metric g on M can be expressed as $f_0^* g_{\mathbb{R}^N} + h$. Use the Nash-Moser iteration to solve the system of PDEs $\partial_i f \cdot \partial_j f = g_{ij}$. The iteration converges by tame estimates on the linearized operator $L_f(h) = df \cdot \nabla h$.

Solution 14:

Stokes: $\int_M d\omega = \int_{\partial M} \omega$ for an $(n-1)$ -form ω . Proof: Use partition of unity to reduce to $\mathbb{R}^n$ case, where it follows from the fundamental theorem of calculus and Fubini. For divergence theorem, let $\omega = \sum_i (-1)^{i-1} F_i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$. Then $d\omega = (\sum_i \partial F_i / \partial x^i) dx^1 \wedge \cdots \wedge dx^n = \text{div } F dV$. The boundary term gives $\int_{\partial M} F \cdot n dS$ by the Hodge star duality.

Solution 15:

For $p = 1$, use $|u(x)| \leq \int_{-\infty}^{x_i} |\partial_i u| dx_i$, multiply over $i = 1, \dots, n$, integrate, apply Hölder repeatedly to get $\|u\|_{n/(n-1)} \leq \prod (\int |\partial_i u|)^{1/n} \leq \frac{1}{n} \sum \int |\partial_i u|$. For general p , apply the $p = 1$ case to $|u|^q$ with suitable q and use Hölder.

Solution 16:

For $u \in C_c^\infty(\mathbb{R}^n)$, write $u(x) - u(y)$ as an integral of ∇u over a chain connecting x and y . Using the mean value integral representation and Hölder inequality, $|u(x) - u(y)| \leq$

$C\|\nabla u\|_{L^p}|x-y|^{1-n/p}$. By density of smooth functions in $W^{1,p}$, the inequality extends, proving the embedding is continuous.

Solution 17:

Suppose u attains an interior maximum at x_0 with $u(x_0) = M > 0$ and $M > \max_{\partial\Omega} u^+$. At x_0 , $\Delta u \leq 0$ (Hessian negative semidefinite), so $Lu = -\Delta u + cu \geq 0 + cM \geq 0$. Strictly, if $c > 0$ or $M > 0$, this gives $Lu > 0$ at the maximum, contradiction. For uniqueness of $\Delta u = f$ with $u|_{\partial\Omega} = g$, difference of two solutions satisfies $\Delta w = 0$, $w|_{\partial\Omega} = 0$, so $w \equiv 0$.

Solution 18:

Cover U by cubes. On each cube where $\|Df\| \leq C$ and $|\det Df| \leq \varepsilon$ on a critical point, use the Whitney extension to show the image can be covered by $O((\varepsilon/C)^m)$ balls of radius $C/\sqrt{n}$. Summing over cubes gives arbitrarily small total measure. The differentiability condition ensures the critical set has enough structure.

Solution 19:

Suppose f has no fixed point. Define $r(x)$ = intersection of ray from $f(x)$ through x with ∂B^n . This gives a smooth retraction (after smooth approximation). By Sard's theorem, regular values exist. For a regular value y , $r^{-1}(y)$ is a 1-manifold with boundary, one endpoint at y . The other endpoint would be in B^n , impossible since r maps interior to boundary. Contradiction.

Solution 20:

Lax-Milgram: Let $a : H \times H \rightarrow \mathbb{R}$ be coercive bilinear continuous on Hilbert H , $f \in H^*$. Then $\exists! u \in H$: $a(u, v) = f(v)$ for all $v \in H$. For the Dirichlet problem, $H = H_0^1(\Omega)$, $a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v$, $f(v) = \int_{\Omega} f v$. Poincaré inequality gives coercivity, trace theorem gives boundary values.

Solution 21:

A one-step method Φ_h is symplectic if $(\Phi_h)^*\omega = \omega$. For Verlet: $q_{n+1/2} = q_n + \frac{h}{2}M^{-1}p_n$, $p_{n+1} = p_n - h\nabla V(q_{n+1/2})$, $q_{n+1} = q_{n+1/2} + \frac{h}{2}M^{-1}p_{n+1}$. Direct computation shows the Jacobian satisfies $J^T\Omega J = \Omega$. By the backward error analysis, the numerical solution is the exact flow of a nearby Hamiltonian, giving $O(e^{-c/h})$ energy conservation over exponentially long times.

Solution 22:

($\Rightarrow$): If A generates $T(t)$, show $R(\lambda, A) = \int_0^\infty e^{-\lambda t}T(t)dt$, so $\|R(\lambda, A)\| \leq \int_0^\infty e^{-\lambda t}\|T(t)\|dt \leq 1/\lambda$. ($\Leftarrow$): Define $A_n = nAR(n, A)$ (Yosida approximants) which are bounded generators of $T_n(t) = e^{tA_n}$. Show $T_n(t)$ converges strongly to a semigroup $T(t)$ and its generator is A .

Solution 23:

The resolvent $(-\Delta + I)^{-1} : L^2 \rightarrow H^2$ is compact by Rellich-Kondrachov. The spectral theorem for compact self-adjoint operators gives discrete eigenvalues $\mu_n \rightarrow 0$, so $\lambda_n = 1/\mu_n - 1 \rightarrow \infty$. Finite multiplicity follows from finite-dimensional eigenspaces of compact operators. The variational characterization follows from the Rayleigh quotient. The condition $\int u = 0$ removes the trivial $\lambda_0 = 0$.

Solution 24:

The Hahn-Banach theorem guarantees the existence of such extensions (Banach limits), but they are not unique and not continuous in the sense of being representable by a sequence. The key issue: ℓ^∞/c_0 is non-separable and the quotient map is not injective. Banach limits exist but require the axiom of choice. They satisfy $\liminf x_n \leq L(x) \leq \limsup x_n$ and are shift-invariant.

Solution 25:

For $F \in \mathfrak{X}(\Omega)$, choose a partition of unity subordinated to a cover by coordinate charts that are half-balls near boundary. In each chart, the divergence theorem reduces to integrating a partial derivative over a cube (or half-cube), which by Fubini and FTC gives boundary integrals. Summing yields $\int_\Omega \operatorname{div} F \, dV = \int_{\partial\Omega} F \cdot n \, dS$. Green's identities follow by setting $F = u \nabla v$.